NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

MID SEMESTER EXAMINATION, Jan-June 2023

B. Tech: Semester-4

Course Name: Artificial Intelligence | Course Code: CS44112

Maximum Time: 2 hours | Max. Marks: 30

Instruction:

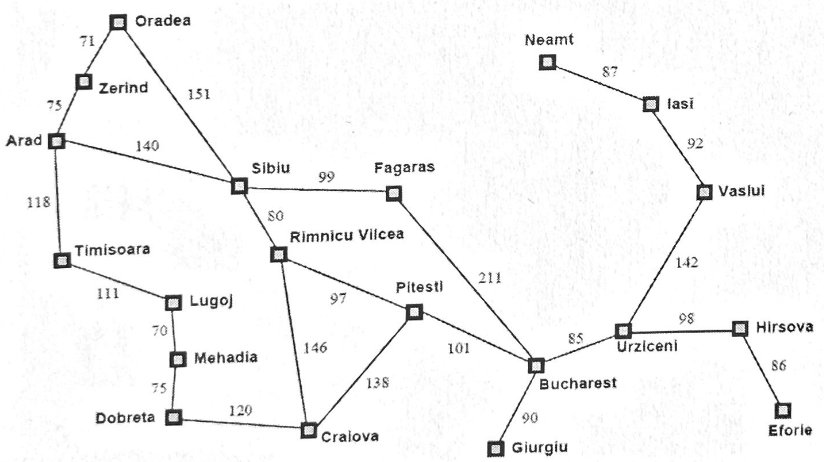
1. Attempt any three questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

1. Find out about the Mars rover. A Mars rover is a motor vehicle designed to travel on the surface of Mars. It is expected to send the types of rocks and water on the surface. — 10 | CO1 | Understanding

a) What are the percepts for this agent?
b) Characterize the operating environment.
c) What are the actions the agent can take?
d) How can one evaluate the performance of the agent?
e) What sort of agent architecture do you think is most suitable for this agent?

2. Find a solution path in the following graph from Arad to Bucharest using the following algorithms. Which solution is optimal? You can consider two parameters i.e., number of states and total cost? — 10 | CO3 | Apply

a) Greedy best-first search algorithm [evaluation function: f(n) = h(n)]
b) A* algorithm [evaluation function: f(n) = h(n) + g(n)]



Straight–line distance to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

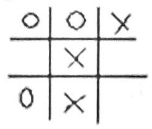
3. Solve the given 8-puzzle using any heuristic method by devising a suitable heuristic function. — 10 | CO3 | Analyze

Start state

1	2	3
8	5	6
4	7	

Goal state

1	2	3
4	5	6
7	8	

4.	Define Production System. Develop a production system for the following problem. "The water jug problem: You are given two jugs, a 4-litre one and a 3-litre one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. Develop a production system to get exactly 2 liters of water into 4-litre jug." [State states, rules and any control strategy]	10	CO2	Understanding
5.	Explain minimax algorithm with an example. Consider the Tic–tac-toe game. Starting from the board position below, expand the complete game tree and calculate the value of each board position. 	10	CO3	Apply